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Preserving medical information from doctor's prescription ensuring relation among the terminology

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ABSTRACT

Healthcare is a fundamental human right, yet accessing proper healthcare remains a significant challenge. Many patients still rely on physical documents, requiring them to carry all relevant medical records during consultations. Existing methods for extracting data from medical prescriptions have primarily focused on recognizing medication names using manual image annotation or binarization techniques. These approaches often fail to capture detailed prescription information, struggle with multilingual text, and lack the ability to structure medicine-related data comprehensively. To address these limitations, we propose an advanced Electronic Health Record (EHR) system that provides a secure and accessible digital platform for storing and managing patients' medical histories. Our study implements the best Region of Interest (ROI) detection model using the You Only Look Once (YOLO) framework, achieving 99.6% detection accuracy. The extracted ROI is processed using the best Optical Character Recognition (OCR) technique to digitize prescription data, which is then organized into a structured format within a central database. Furthermore, the system integrates a spell correction algorithm with a 96% accuracy rate to rectify misspelled medication names. Beyond text extraction, the system links corrected medication names with dosages, instructions, and manufacturer information, combining generic and brand details to ensure precise and comprehensive healthcare data management. This integrated solution enhances medication data organization, facilitates better healthcare delivery, and improves patient outcomes by streamlining prescription handling and ensuring accurate medication administration. Our EHR system bridges the gap between physical and digital records, advancing an efficient and reliable healthcare ecosystem.

1. Introduction

Health is a complete harmony of the body, mind, and spirit. Countries face challenges managing their healthcare systems due to their high-density populations, poor record keeping, staff shortages, and mismanagement [1]. Bangladesh is a developing country in South Asia, with about 160 million people, according to recent estimates, and it too faces many challenges in its healthcare sector [2,3]. Several factors can explain why countries like Bangladesh's medical infrastructure is so fragile and complex [4,5]. Although the governments of many countries around the world are investing in improving their medical infrastructure. In 2021, during the second year of the COVID-19 pandemic, countries around the world spent more on healthcare, reaching a total of US$ 9.8 trillion, which is 10.3% of the global economy [6]. For example, the budget of the Government of Bangladesh in the fiscal year 2022–23 was 6780.64 billion and increased by 12.34% in the following year to improve the health sector [7]. When someone sees a medical professional, she/he has to go through some tedious procedure. But in many countries patient's medical history is absent in a central repository of records, patients have to bear hard copies of all their medical documents whilst visiting a doctor [8,9]. Previous medical history is essential for proper diagnosis of the disease because a doctor must be aware of a patient's past health issues, prescribed medication, test reports, etc. All these procedures are time-consuming and there are chances that the hard-copied documents are lost and due to that a patient's diagnosis can be affected massively. But what if a patient's past medical history is documented digitally. An Electronic

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Health Record (EHR) is a digital version of a patient's medical documents containing extensive information about an individual's medical history and other necessary information [10,11]. In a structured medical system EHR plays a vital role whilst treating a patient [12,13]. Due to its remote access to information, streamlined documentation, and systematic structure, EHR is efficient and recommendable for a patient [14–16]. Many countries have their own EHR systems such as My Health Record is Australia's secure online summary of health information. The Canadian provinces have separate EHR systems. The United States uses a decentralized EHR system. Turkey's e-Nabız system allows people to manage their health via web and mobile applications. Hungary's EESZT system stores and shares medical documents, and is evolving to transform data into structured formats [17]. Despite advances, implementing EHR systems remains costly, particularly for small or independent healthcare practices, with maintenance costs per physician previously estimated at $1500 per month, based on decade-old data from the era of on-premise systems [18]. More recently, 39% of independent practices identified cost management and financial concerns as primary challenges in adopting EHR systems [19]. Healthcare facilities in developing countries primarily use handwritten and physical prescriptions, leading to data loss, misunderstanding, and negative patient care. The absence of standardized EHRs and low adoption of technologies further exacerbate these issues. Digitalization aims to optimize healthcare systems, reduce medical errors, and improve treatment outcomes. In developing countries like Bangladesh patients often face the challenge of preserving and managing all their medical documents [20]. An EHR system can significantly mitigate these issues by enabling patients to collect and maintain all their medical documents in a centralized digital format [21]. Healthcare sectors face additional challenges in managing data from medical papers, particularly due to the difficulty of deciphering crucial information from doctors' often cursive handwriting [22,23]. Recognizing handwritten medical prescriptions is much more complicated than recognizing regular handwritten documents because of the messy writing style of doctors and the usage of less common medical terms. Because the terms are not seen in normal text, special models are required to understand them [24–26]. Prescriptions do not follow a standardized format, and consistent terminologies, making it harder to use common handwriting recognition methods [27]. These issues showcase the necessity for specialized solutions to read prescriptions accurately. Clinical documents are noisy and full of medical language, and misspellings and grammatical errors [28]. Effective medical treatment relies heavily on the availability of a patient's comprehensive medical history. To address this, there is a need to develop a centralized or local database system that can store and retrieve a patient's previous medical records efficiently. Prescriptions also have more than one section each of these sections may contain critical information that needs to be accurately extracted and structured for effective use. Prescriptions in many countries often contain information in multiple languages, adding another layer of complexity to the extraction process. Some previous studies have demonstrated solutions for enhancing healthcare interoperability. For example, the Structure Mapping mechanism changes different types of healthcare data into a standard format called Health Level 7 (HL7) Fast Healthcare Interoperability Resources (FHIR) standards using ontology alignment [29]. Another approach organizes data from medical devices (IoMT) into HL7 FHIR using 5G networks for quick processing [30]. A Personal Health Record app allows easy two-way data sharing between patients and EHR systems, making healthcare more efficient [31].

Previous medical prescription data extraction systems mainly focused on extracting medicine names from physical prescriptions [32–34], but they were unable to capture other important details of the prescription. Additionally, some systems have achieved high accuracy in medicine name recognition, but are limited to recognizing names with fewer than 20 characters [33]. In addition, many of the existing systems use the process of manual image annotation or binarization for the detection of the Region of Interest (ROI) area in prescription images [35–37]. Such methods cannot identify all relevant medicine-related data simultaneously, resulting in incomplete information extraction. For instance, some systems manually insert text on a line-by-line basis to overcome segmentation challenges [36], showing the limitations of their approach to automated processing. However, some of the methods developed in this approach can quite well identify and classify medical text from prescription prescriptions without preserving context [35], which limits their application. Furthermore, the storage and management of extracted data have been neglected in previous studies, with no clear way and purpose of how the extracted data can be used or archived [35]. These limitations prevent the creation of a comprehensive system for extracting prescription data.

To address these challenges, the authors of this paper propose a new method for identifying and extracting medicine-related data from both handwritten and printed medical prescriptions in multiple languages. The specific contributions are:

- In this work, a YOLO-based Region of Interest (ROI) detection model is proposed, achieving 99.6% accuracy and significantly outperforming existing models in accurately identifying key areas in both handwritten and printed prescriptions across multiple languages.
- The experiments have been conducted using an integrated system combining Optical Character Recognition (OCR) and spelling correction algorithms, achieving 96% accuracy in extracting and correcting medication-related information, thus improving upon the limitations of current OCR-based methods.
- Finally, a comprehensive framework for digitizing and structuring medical records enhances long-term patient care by providing healthcare professionals with reliable and accessible patient histories, outperforming traditional digitization techniques.

The rest of the paper is organized in the following manner: literature review discussed in Section 2. Section 3 reflects an overview of our diverse dataset. In Section 4 a detailed breakdown of the proposed system is provided. Evaluation and results of this research and comparisons can be found in Section 5. The conclusion can be found in Section 6. The future work can be found in Section 7.

2. Literature review

Text is the most important object among all the other objects present within an image. The text makes work easier for both humans and machines. Because it provides rich, precise, high-level information. Nowadays, the text classification technique is being used in the medical domain. We can use medical documents, after localizing the text regions we can extract information from the scanned/captured image for the classification procedure.

2.1. Text localization

Dhande et al. [32] focused on character recognition for English cursive handwriting. Their work is dedicated to recognizing medicine names from doctor's prescriptions. They proposed a horizontal projection method for text-line segmentation and a histogram method used for word segmentation. 125 features were extracted from each character using the convex hull algorithm and SVM was used for classification, on 50 different prescription samples achieving 95% accuracy for text-line segmentation, and 92% for word segmentation. To improve the accuracy, the recognized word has been matched with a dictionary and came up with 85% recognition accuracy of the system. Their work is only limited to medicine names.

Edlabadkar et al. [33] proposed a system to recognize medicine names on handwritten prescriptions. Visually impaired individuals can benefit from this system as they have converted medicine names into audio format. Two distinct models: YOLOv4 and CRNN-CTC, are used for prescription region identification medicine name, and strength

recognition. Google Text-to-Speech module implemented for audio conversion. They have achieved approximately 98% accuracy rate and a character error rate of 0.0029. Their model cannot recognize doses and medicine names with more than 20 characters.

2.2. Medical document analysis

Sarkar et al. [38] explored word spotting in cursive handwritten documents by applying a modified character shape code technique. It introduces a two-level selection process, leveraging word size and character shape code, to enhance the efficiency of matching search queries. The proposed method, an adaptation of the Modified Character Shape Code algorithm, is implemented for word spotting in cursive handwritten document images. Additionally, the method exhibits reduced accuracy when applied to search query words with four or fewer characters.

Roy et al. [39] introduced keyword-based searching in prescription to efficiently extract relevant data. They divided a prescription into two parts, printed letterhead and the handwritten part or body. 500 prescriptions had been trained using the Hidden Markov Model, and they achieved encouraging results. They proposed a multilayer perceptron-based Tandem feature to improve the performance. However, the search efficiency entirely depends on the size of the keyword database.

Shanmugalingam et al. [40] addressed the need for automatic language identification in the context of social media, particularly for code-mixed text. The paper proposes a novel approach using Natural Language Processing (NLP) and Machine Learning (ML) techniques to provide word-level language tags for Tamil–English code-mixed data from social media posts and comments. The paper achieved a high accuracy of 89.46% in language identification using the SVM classifier for Tamil–English code-mixed data in social media.

Carchiolo et al. [41] researched to extract text from scanned prescriptions and effectively classify medical prescription data using Natural Language Processing (NLP) and machine learning techniques. It employs a RESTful Web Service and uses syntactic rules and rule-based tagging. First, it shows that 30% of prescriptions are unclassifiable, especially when spelling corrections are not needed. This improved solution significantly improved classification, with only 5% of prescriptions considered unclassifiable.

Achkar et al. [36] used Convolutional Recurrent Neural Network (CRNN) to reduce human error in interpreting medical prescriptions. They acknowledge the challenge of deciphering doctors' handwritten prescriptions, which often contain Latin abbreviations and medical terminology. The proposed network consists of a stack of 13 convolutional layers followed by three Bidirectional LSTM layers. They have achieved the best recognition at the line level. In the case of an image with multiple prescriptions, their system is not able to detect the start and end of the prescription. The problem faced is the division of the images of each prescription into small segments. They manually inserted text on a line-by-line basis to overcome this problem. Their system achieved a good recognition rate with 98% accuracy.

Khandokar et al. [42] applied handwritten character recognition (HCR) using a convolutional neural network (CNN). The process of image processing involves pre-processing, segmentation, feature extraction, classification, and recognition. Feature extraction is done using a CNN with a ReLU activation function, which forms a reduced matrix and compacts it into a feature vector. The accuracy of the CNN improves with increasing training images, reaching 92.91%. The system trained with 1000 images from the NIST dataset, achieving 92.91% accuracy with 200 testing images. The accuracy of training images cannot exceed the 92.91% limit due to numerical errors and constraints on CNN image differentiation for labels.

Tabassum et al. [43] Created a system that can recognize a doctor's handwriting and generate digital prescriptions. A bidirectional Long Short-Term Memory (LSTM) network is used in the study as a machine learning method. To predict doctors' handwriting, an online character recognition system was developed using a bidirectional LSTM model. Using bidirectional LSTM and RSS data augmentation, this study conducted eight trials on the handwritten data set and achieved an average accuracy of 93.0% (maximum: 94.5%, minimum: 92.1%). Compared to the recognition results without any data expansion, this accuracy was 19.6% higher. Handwritten recognition technology in smartpens can digitize doctors' writings in real time.

Sakib et al. [35] extracts and classifies medical text from prescription photographs. Their proposed system works in four phases: text localization and extraction from prescription photographs, image classification, image-to-text conversion using Optical Character Recognition (OCR), and text classification into different categories e.g., symptoms, medications, investigation, and others. For localized image classification VGG-16 has been used and Bidirectional Encoder Representations from Transformers (BERT) model for text classification. For image classification and medical text classification, their model obtained an accuracy of 98.37%, 99.9%, and 95.93%, 99.4% for training and test sets respectively. The proposed system seeks to establish a medical history archive while extracting every word in an unordered manner.

Pavithiran et al. [37] created an application that helps the majority of user to verify the doctor's handwritten notes or prescriptions without any prior knowledge of calligraphy analysis and it delivers in the language of their choice. Deep learning methods including CNN, RNN, and LSTM are used to train the model for recognizing doctors' handwritten prescriptions while Unicode was used to match words from various languages. The model was trained with 50 epochs divided into 90% for training and 10% for testing and kept low on the CTC loss function to get the best possible word prediction. Fuzzy search and market basket analysis were used to optimize the results using a pharmaceutical database.

2.3. Text extraction from images

Paul et al. [34] extracted medication names from handwritten prescriptions using a "weakly supervised" method. The paper introduces a weakly supervised method for extracting medicine names from handwritten prescriptions using weakly labeled data. It uses weak labels to identify regions of interest and injects a domain-specific medicine language model(n-gram model). The model uses a pre-trained OCR backbone and a CTC decoder for symbol classification. The suggested method outperforms existing methods for extracting drug names from handwritten prescriptions using weakly labeled data by more than 2.5 times.

Sharma et al. [44] discussed the use of OCR systems to extract text from photos and convert it to ASCII text. The process involves digitization, pre-processing, text localization, splitting, categorization, and character recognition. The system uses Tensorflow and Keras for data pre-processing and the accuracy of CNN classification of handwritten digits is evaluated using a single layer and a hidden layer. The paper also mentions the Leptonica library, an open-source software for computer vision and image processing applications. The maximum accuracy is found to be on average, 93%. The MNIST dataset is used as a training sample by the system. This project reduces the complexities and improves the accuracy of OCR. This system can achieve an accuracy of around 93% on average, the highest it has ever achieved.

Vedhaviyassh et al. [45] addressed the application for license plate recognition in various cases. This highlights the practical significance of the research in developing automated systems that are required for vehicle detection. The author proposes a structured approach consisting of three main modules: image acquisition which involves vehicle photography, license plate detection using the YOLOv5 model and the final step, the paper uses OCR methods, specifically Tesseract OCR and EasyOCR, To recognize and extract characters from detected license plates. A dataset obtained from GitHub is used to train and test the proposed model. The results reveal that EasyOCR outperforms Tesseract

Table 1
Comparison of existing literature and advancements in prescription data extraction.

Author	Focus	Strengths	Limitations	Advancements in this work
Dhande et al. [32]	Recognizing medicine names from prescriptions	Effective segmentation and dictionary matching improved recognition accuracy	Limited to medicine names; does not capture other critical information from prescription	Extracts comprehensive prescription information, including dosage, form, etc., and employs spell correction for improved accuracy.
Edlabadkar et al. [33]	Automated medicine name recognition and audio conversion	High accuracy (98%); automated recognition reduces errors	Cannot process medicine names exceeding 20 characters; lacks other information recognition	Processes all medicine-related data; achieves 99.6% ROI detection and robust spell correction
Achkar et al. [36]	Line-level recognition, resolving human error in interpreting prescriptions	High recognition accuracy (98%); robust model for line-level recognition	Unable to detect start and end of prescription, requires manual insertion of text	Accurate segmentation and processing for all prescription fields
Sakib et al. [35]	Extracting and classifying text from prescription photos	High accuracy (99.9% for text classification); builds a medical history archive	Unordered word extraction; no relationship or sequence between words	Structured and sequenced extraction, ensuring relationships in prescription data
Pavithiran et al. [37]	Multilingual prescription recognition with deep learning and fuzzy search	Effective word matching using fuzzy search; multilingual support	Small training/testing split; focus limited to word prediction, Limited to English handwritten and printed text	Expanded to include multiple languages, large-scale dataset; robust spell correction; centralized structured data storage
Paul et al. [34]	Extracting medicine names with weakly labeled data	Improved medicine name extraction	OCR errors hinder advanced extraction; Relies on weakly labeled data, limited to extracting medicine names	High spell correction accuracy; Improved accuracy and ability to extract medicine-related data, not limited to weakly labeled data

OCR, achieving 95% accuracy in predicting license plates, compared to Tesseract's 90% accuracy. This suggests that EasyOCR's deep learning approach is more effective for real-time predictions in license plate recognition tasks.

2.4. Spell checking and correction

Yulianto et al. [46] focused on autocomplete and spell-checking using the Levenshtein distance algorithm. This paper's objective was to increase the accuracy of text suggestions in searching library data. This paper introduces the Levenshtein distance algorithm for text suggestion and spell-checking. The study uses a dataset of 1155 data entries obtained from the UNNES library, which serves as the basis for evaluating autocomplete and spell-check features implemented with the Levenshtein distance algorithm. The dataset includes information on input processes and the classification of books. The accuracy of the algorithm is evaluated through data search trials and the performance of the Levenshtein algorithm was evaluated based on 1055 source cases and 100 target cases, resulting in an 86% accuracy rate in detecting and suggesting text corrections during the search process.

2.4.1. Healthcare data interoperability and standardization

Lete et al. [47] introduced the Terminology Service as part of the CrowdHEALTH project, which helps healthcare systems work together by converting local codes into standard ones like ICD-10 and SNOMED-CT. It uses HL7 FHIR standards and the project includes a rule engine for business logic implementation, a Structure Mapping Service for knowledge creation and management, and a Terminology Service for operations on medical terminologies. The CrowdHEALTH project provides an infrastructure for converting clinical information into meaningful data, enhancing healthcare system communication. Future work focuses on extending and testing the system further.

Kouremenou et al. [48] explored how to improve data exchange and system interoperability in healthcare through the use of the Fast Healthcare Interoperability Resources (FHIR) standard. This process defines those elements of key data in the process and links each of them to the appropriate FHIR resource, and sets up relationships and search options. FHIR provides the means for the smooth data sharing through use of the global standards, such as ICD-10 and SNOMED-CT. One of their methods makes for an easy data sharing and communication for non technical users between healthcare providers. The study also identifies obstacles like the absence of practical guides and provides future work that merges AI with other fields to promote its use in cultural heritage protection.

A summary of the existing literature and the proposed work is presented in Table 1. The existing studies have been categorized based on the focus area, strengths, limitations, and advancements in the methodology used for prescription data extraction. This comparison highlights the key contributions of each work while emphasizing the improvements introduced in the proposed approach.

3. Dataset description

In our research, we developed multiple datasets to facilitate the analysis and management of medical prescriptions and related information. These datasets are crucial for implementing and testing various components of our proposed system, including prescription data extraction, medicine identification, and symptom cross-checking. The datasets include a variety of information from prescriptions, annotated images, medication details, and disease-specific data, all meticulously gathered and organized to support our objectives. An important aspect of healthcare data analysis is data quality. Healthcare data may have issues like missing information, data quality, and irrelevant data as it comes from different sources and methods [49] which also hinder accuracy. Therefore, careful data preprocessing techniques are needed to improve accuracy and performance in the healthcare sector [50].

3.1. Prescription dataset

In Bangladesh, hospitals, clinics, and pharmacies do not have an organized structure for storing prescription data. When a patient visits a hospital or clinic, the doctor prescribes medication on physical prescriptions that are not retained by the healthcare facilities. Instead, patients are responsible for keeping their prescription records. While some large private hospitals may store patient prescription data, this is not a universal practice. Furthermore, pharmacies in Bangladesh traditionally dispense medications based on physical prescriptions presented by patients and typically do not retain prescription data. As a result, prescription data in Bangladesh is collected directly from patients. Because of this, we collected prescriptions from our friends and family only after getting their permission. Before collecting any

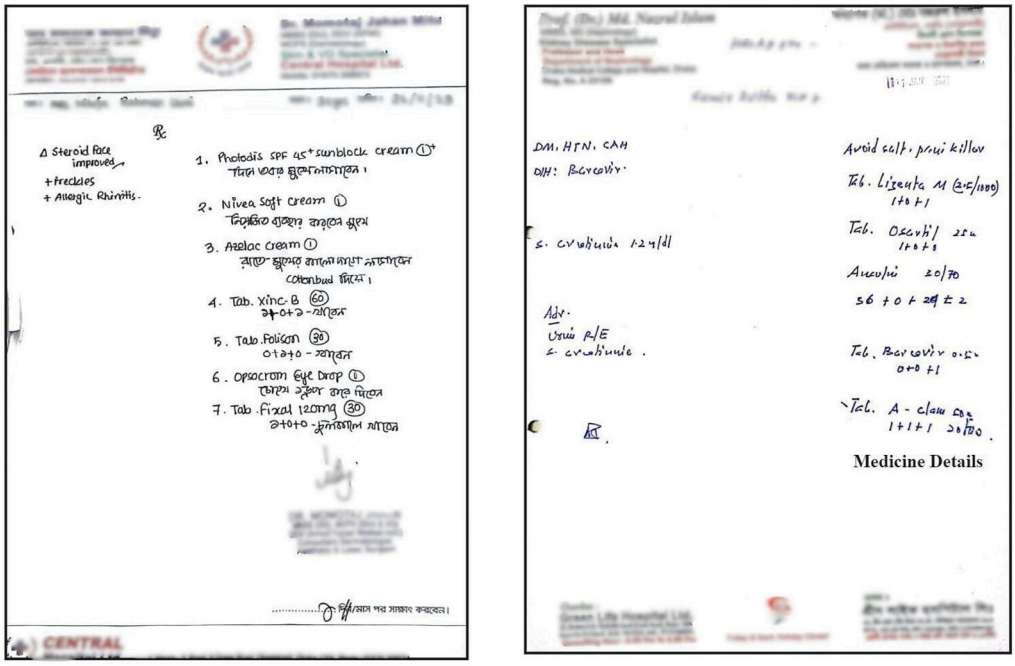


Fig. 1. Sample prescriptions commonly used by healthcare professionals.

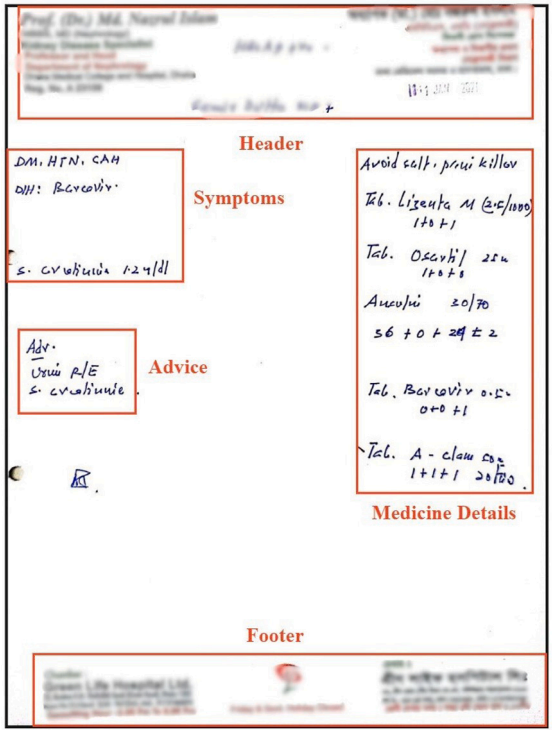


Fig. 2. Sample prescriptions commonly used by healthcare professionals.

data, we provided a data collection form to each participant that included some questions to inform participants about the research and its implications. Each participant provided signed consent, acknowledging they had no objections to their anonymized data being used in our research. Prescriptions are typically structured into sections: the header (doctor, patient, and hospital information), body (medicine details, symptoms, and advice), and footer (hospital information). To protect patient privacy, we blurred the header sections of prescriptions, which typically contain personally identifiable information (e.g., patient's name, age, contact details). This research focused on the body section of the prescriptions, which contains medicine-related data. We collected 400 medical prescription datasets, 75% handwritten and 25% printed, with 90% being bilingual (Bangla and English) and 10% exclusively in English. These prescriptions represent diverse handwriting styles and formats from urban and rural regions in Bangladesh, including private and government hospitals and clinics prescriptions. These prescriptions were authentic and real-time, issued by doctors during patient treatments across various hospitals and clinics in Bangladesh with the hospitals and clinics information in the header and footer sections ensuring data validity. To further ensure the validity of our prescription dataset, all collected prescriptions were cross-validated by a medical professional with a Bachelor of Medicine and Bachelor of Surgery (MBBS) degree. This additional step reinforced the reliability of our data collection process. Fig. 1 shows some Prescriptions Samples.

3.2. Localized image dataset

Fig. 2 shows the classification of prescriptions. Here we annotate the prescription image medicine details related part for localized medicine-related data. We use the image annotation technique to detect ROI parts from prescription images using labelimg tool. So we can extract the related data of any medicine together, after annotation. This dataset helps us to train our YOLO ROI detection or localize models to detect or localize our necessary ROI parts from prescription images. Fig. 3, shows the localized medicine-related data from the prescription image. So, we get almost 2200 coordinates of the ROI localization part from the 400 annotated prescription images which are basically in text files.

3.3. Medication information dataset

Standard databases for medication information in EHR databases guarantee data consistency, reduce errors, and improve accuracy. This study developed a medicine information database from a single source through web scraping from MedEx which is a reliable website that contains information that has been verified by professionals. The MedEx website is a trusted source in the medical field, and healthcare professionals widely use its data. In the future to enhance standardization, researchers will pay attention to the incorporation of universal drug code systems.

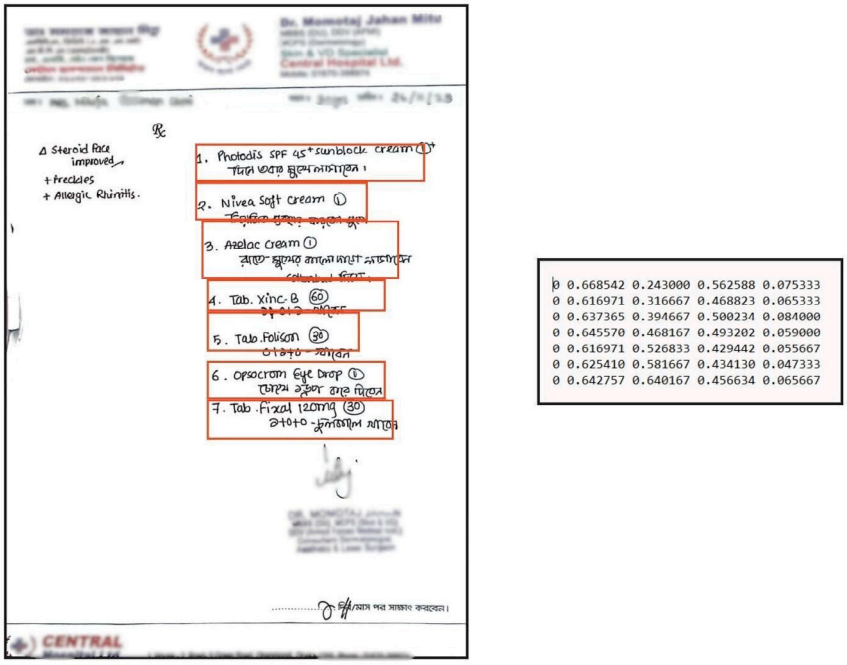


Fig. 3. Region of interest localization of medicine related data from annotated prescription images.

Table 2
Example of some medicine and its corresponding generic name and manufacturer name information.

Medicine name	Generic name	Manufacturer name
5-Fluril	Fluorouracil	Techno Drugs Ltd.
A-Cold	Bromhexine Hydrochloride	ACME Laboratories Ltd.
Adiva	Efavirenz	Square Pharmaceuticals Ltd.
Adlina	Linagliptin	UniMed UniHealth
Adlock	Propranolol Hydrochloride	Sonear Laboratories Ltd.

3.3.1. Medicine related information dataset

Our medicine dataset was created through an extensive web scraping process from a well-established medicine website. This dataset contains vital information on various medications, including their brand names and corresponding generic names. With a total of 15,106 entries, this dataset serves as a critical resource for identifying generic names and finding alternate brands for medications, which is particularly useful for healthcare providers and patients seeking cost-effective or available alternatives. Dataset Structure: The dataset is organized into two main columns: Medicine Name, Generic Name and Manufacturer Name.

Table 2 illustrates examples of the dataset entries, including medicine names, generic names, and manufacturer names. Medicine Name column contains lists of names for various medications. Generic Name column provides the generic names associated with the respective brand names listed in the Medicine Name column. Manufacturer Name column provides the manufacturer names associated with the respective brand names listed in the Medicine Name column.

3.3.2. Indication details dataset

Leveraging advanced web scraping techniques, we have curated an extensive dataset from a reputable medicine website. This dataset encompasses detailed information on various generic medications and the symptoms and diseases they are commonly prescribed to treat. With 1628 entries, this dataset serves as an invaluable resource for healthcare professionals, researchers, and developers engaged in medical applications related to disease treatment and medication management. Dataset Structure: The dataset is organized into two primary columns: Generic Name, and Additional Info. Table 3 provides examples of the dataset entries, illustrating generic names and indication details. Generic Name column lists the generic names of medications or formulations. Additional Info column provides detailed descriptions of the indications, benefits, and usage instructions for each medication.

Table 3
Example of some medicines and their corresponding additional information.

Generic Name	Additional Info
Abiraterone Acetate	Abiraterone Acetate is used with prednisone to treat metastatic prostate cancer.
Acalabrutinib	Acalabrutinib is a kinase inhibitor for treating Mantle Cell Lymphoma (MCL), Chronic Lymphocytic Leukemia (CLL), and Small Lymphocytic Lymphoma (SLL).
Acarbose	Acarbose helps improve glycemic control in adults with type 2 diabetes, alongside diet and exercise.
Aceclofenac	Aceclofenac is used for pain and inflammation relief in conditions like arthritis, toothache, trauma, and lumbago.
Acemetacin	Acemetacin treats pain and inflammation related to joint disorders, such as rheumatoid arthritis and postoperative pain.
Acetate Formulation	Acetate Formulation is used for hemodialysis.

4. Methodology

In this work, we proposed a new method for identifying, and extracting medicine-related data from medical prescriptions. Prescriptions are available in printed and handwritten in multiple languages and may contain more information than the necessary details. This method extracted the necessary data from the prescription. Next, it helps to store all the information in a structured format. This system also proposed a method that helps to spell the medicine name correctly, give any medicine all available brand name details with their company name, and also give any medicine name indications and store those pieces of information in a structured format. Fig. 4 outlines the process of extracting, correcting, and storing medical data from prescription images, including medication details via OCR and web scraping.

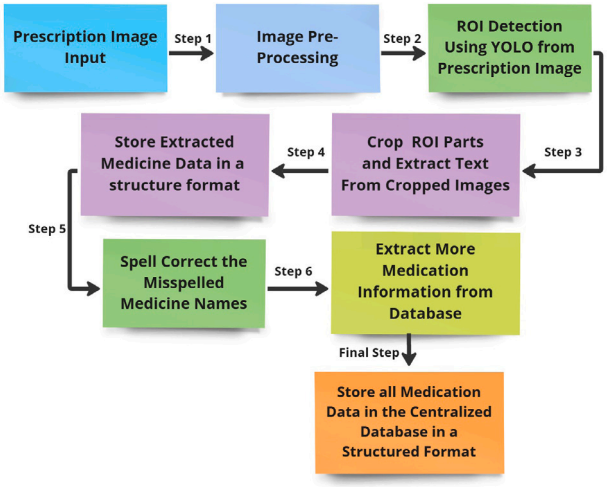


Fig. 4. Medical data extraction workflow.

4.1. Workflow of the proposed system

The main purpose of the system is to help people maintain medical records easily and to provide people with information about the use of medicine and other related brands. If the user provides a prescription image in the system, it localizes the medicine-related parts from there and then uses OCR to extract the medicine-related text from the image and store it in a structured format. Then spell correction method corrects the name of the medicine if it is wrong. Then the system extracts more medication information. Figs. 5 and 6 shows the overview of the whole proposed system which is divided into two parts.

As Fig. 5 shows, the proposed system uses image pre-processing and annotation techniques to identify ROI in prescription images. It then automatically detects ROI parts from the prescription image using the YOLO ROI detection model which is trained by the annotated prescription images. After doing this the system uses a cropping algorithm to crop the ROI localization parts from the prescription image. The system then uses OCR technology to extract text from the cropped image and stores the extracted text in a well-structured format.

As Fig. 6 shows, the proposed system uses a spelling correction method (Spello model, Levenshtein edit distance algorithm) to correct any errors in the extracted medication names. In addition, it uses the medicine name from the structure data to get more medication information using the obtained dataset from web scraping. Finally, store those extracted data in a centralized database.

4.2. Image pre-processing

The proposed system requires high-quality prescription images for effective training and testing. However, the image quality may suffer due to device differences, lack of lighting, hand-clipping dust, etc during data collection. So the proposed system uses noise reduction and image sharpening techniques to fix the image quality to increase the accuracy and performance of the proposed system steps during training and testing.

4.3. Prescription image annotation for medicine data extraction

Image annotation is the labeling of objects of interest within images or videos in the context of object detection in YOLO models. The proposed system annotates the collected prescription images using the LabelImg tool. After annotation, we get the axis of the ROI localization part from the annotated image. Here the proposed system annotates the prescription image medicine details related part for localized medicine-related data. So the proposed system can extract the related data of any medicine together, after annotation. When the model is trained, the model will be able to identify the boxes that surround these medicine-related data and tell the location and type of the data. The steps of the annotation process are shown in Algorithm 1.

Algorithm 1 Annotating Procedure

Input: List of images to be processed.
Output: Text files containing normalized coordinates and class information for each image.

```
ImgList ← getAllImgInDir(path)
while ImgList is not empty do
    img ← ImgList.pop()
    class_no ← getClassNo()
    x, y, w, h ← getCoordinates(Draw(img))
        [x and y is the top-left corner of the bounding box & w and h
    is the width and height of the bounding box]
    Normalize coordinates:
        x ← x / img_w
        y ← y / img_h
        w ← w / img_w
        h ← h / img_h
    write(class_no, x, y, w, h) to file
end while
```

4.4. ROI localization and detection model

In this phase, the proposed system does training and evaluating YOLO ROI detection models (YOLOv7, YOLOv7 Tiny, YOLOv8, and YOLOv5) [51–53] on custom annotated prescription images, Annotated prescription images are used as the input data for training the detection models. Annotations typically include bounding box coordinates and class labels for regions of interest in the images. YOLO is a neural network designed to predict ROI localization from images and videos. It divides an image into s*s grid. Each grid cell predicts B number of bounding boxes with the dimensions of the box and the confidence score which indicate how confident the model is that the object is inside the bounding box. We define confidence as

$$\text{Confidence Score} = \Pr(\text{Object}) \times \text{IOU}_{\text{pred, truth}} \quad (1)$$

Pr (Object) = Object Probability
IoU = Bounding Box Probability

In Eq. (1) here object means ROI localization parts of the prescription image and pr(object) means whether there is an object or not which is expressed by 0,1. IoU means intersection over the union between predicted and ground truth bounding box. A higher IoU means that the predicted coordinates for the bounding box are close to the coordinates of the actual box in the ground truth [54].

Fig. 7 shows how well an object detection algorithm will detect the objects in the image. The YOLO ROI localization models divide the prescription image into grids. The image is processed by CNN for feature extraction. CNN architecture like Draknet or Resnet is mainly used here. It basically generates multiple bounding boxes for each grid cell, confidence score, and class probability. Here, non-maximum suppression is used to fix multiple bounding box detection problems by selecting the box containing the highest confidence score for primary detection.

Here the proposed system trains YOLOv7, YOLOv7 Tiny, YOLOv8, and YOLOv5 models for ROI detection using prescription images. After training, the performance of each model is evaluated using performance metrics such as precision, recall, and mAP (mean Average Precision). The results from YOLO models are analyzed to identify the best model YOLOV7 model which is then used in the test phase to draw bounding boxes around prescription images to locate and identify regions of interest, providing bounding box coordinates, confidence scores, and detected object class labels as an output.

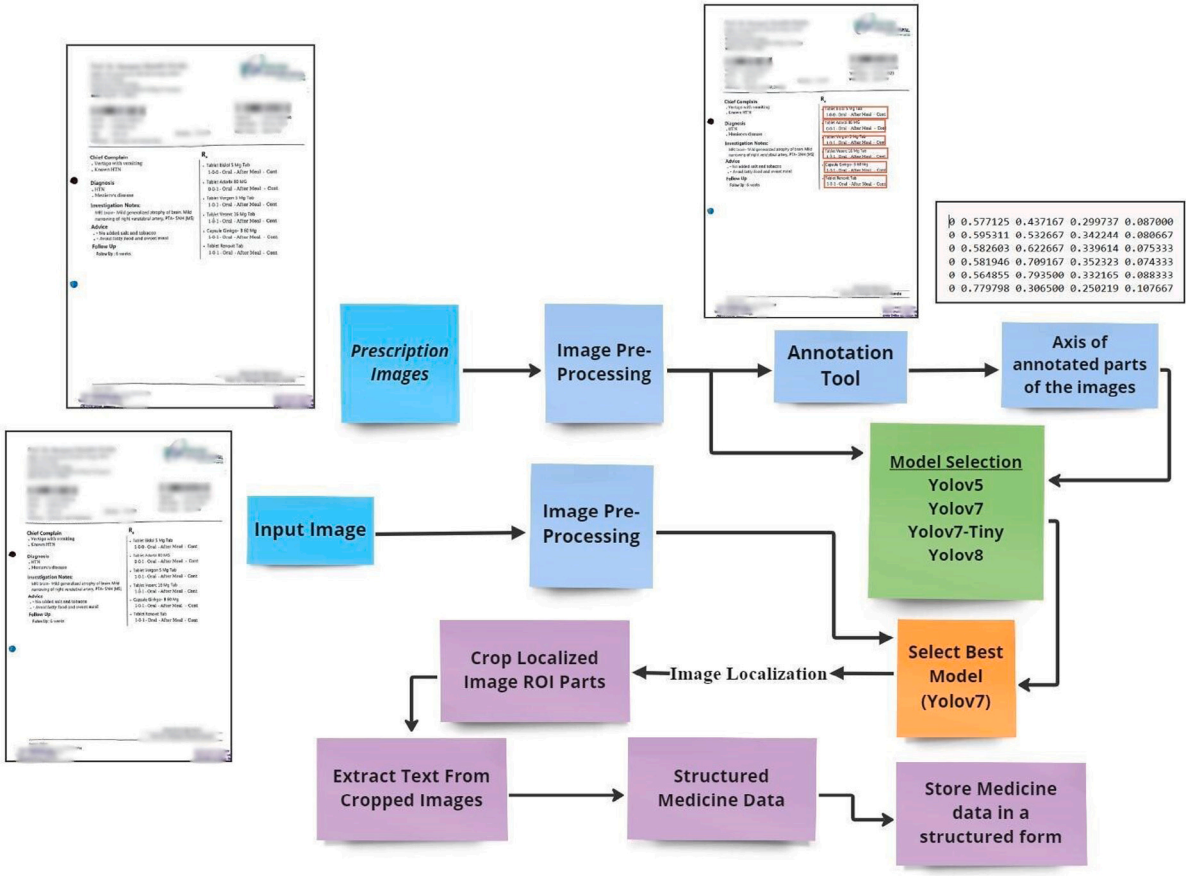


Fig. 5. Text extraction and storing process of prescription information.

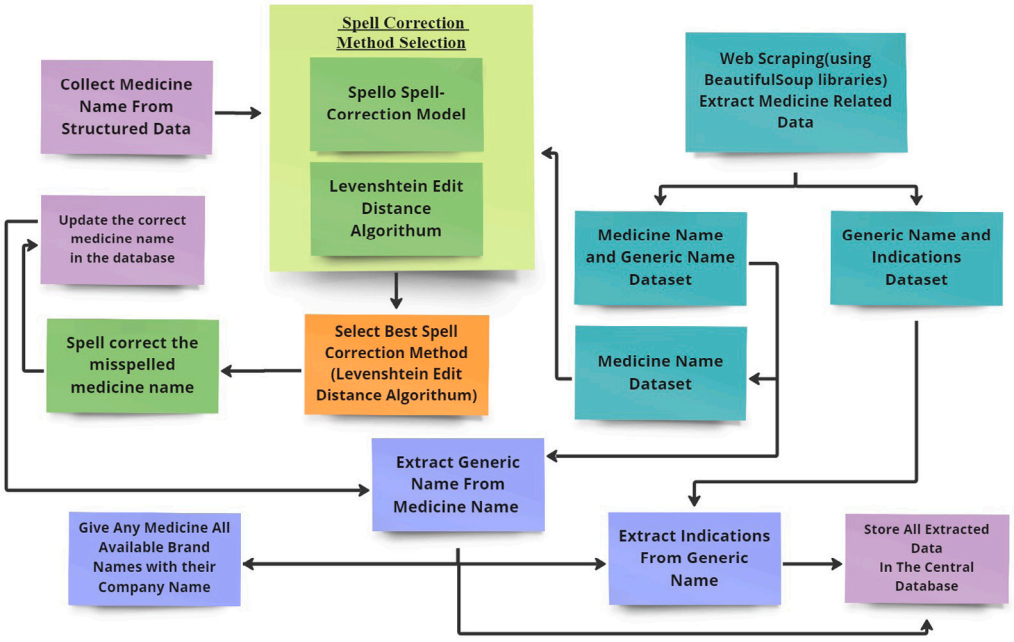


Fig. 6. Process of spell correction and medication information extraction.

4.5. Image cropping based on localized images

This step of the proposed system uses a cropping algorithm that can automatically extract and preserve predefined regions of interest from an image. First, the proposed system gives an image as input to the algorithm and assumes the axis of the annotated parts of the image, which contains the class labels of the ROI localization parts and information about the center point, width, and height of the possible rectangular area (bounding box) that surrounds the original ROI localization part. That is inside the ROI bounding box. The algorithm reads

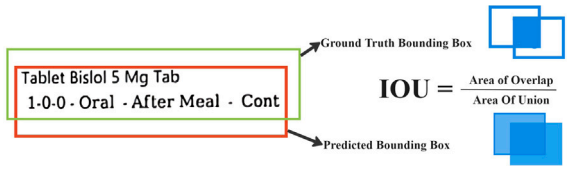


Fig. 7. IoU calculation technique.

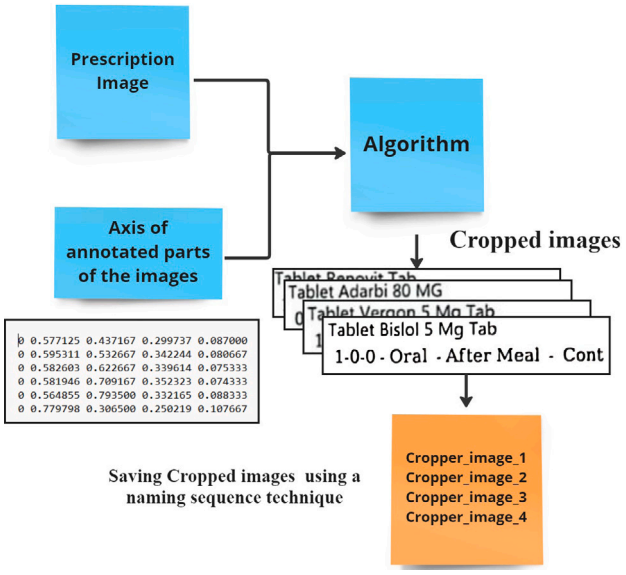


Fig. 8. Extraction and saving of localized ROIs from prescription images.

the image and text file to iterate through all the axes of the annotated parts of the image. For each axis of the annotated segments, it calculates the relative coordinates of the center of the image and the exact pixel coordinates of its upper left corner based on the width and height information to convert the relative values to actual pixels. Finally, the code stores all extracted annotated segments as individual image files in temporary storage. Save files follow a naming pattern for further use where the base name "cropped_image_" is used with a unique number for each saved image, eg, cropped_image_1.jpg, cropped_image_2.jpg, etc.

Fig. 8 shows the extraction process of localized ROIs from prescription images and saves them as separate images. The process of Cropping ROI images from the prescription images is shown in Algorithm 2.

4.6. Text extraction from cropped images

At this stage, the proposed system extracts text from the ROI-localized image. The proposed system uses ROI-localized images as input with their unique filenames (eg, cropped_image_1.jpg). Now the algorithm iterates through all the images and it extracts the set number from the filename using the regular expression. If the file name does not follow the pattern, the image will be omitted. Then the proposed system extracts text from the Prescription ROI local image using EasyOCR [55] and Tesseract OCR [56] for multiple languages. EasyOCR gives the most suitable result because EasyOCR is more accurate than Tesseract OCR as it uses deep learning techniques to recognize objects and has high efficiency in real-time prediction. It subsequently prints the extracted text and saves it in a text file with the name "text{set_number}.txt" in temporary storage, which captures every set of text. For instance, the extracted text from cropped_image_0 is saved in a text file named text1.txt, while that from cropped_image_1 is saved in a file named text2.txt and the other files to follow analysis or downstream tasks. The prescription ROI images text extraction process is shown in Algorithm 3.

Algorithm 2 Bounding Box and Cropped Image Extraction

Input: PrescriptionImage, image[] and AnnotationFile, file[] **Output:** cropped_image[]

```
image_height, image_width = image_shape
lines[] = file[]
NumberOfBounding_boxes = NumberOfLines(lines[])
for i = 1 to NumberOfBounding_boxes do
    x_center_pixel = lines[i].x_center × image_width
    y_center_pixel = lines[i].y_center × image_height
    box_width = lines[i].width × image_width
    box_height = lines[i].height × image_height
    x = x_center_pixel - box_width / 2
    y = y_center_pixel - box_height / 2
    bounding_boxes(x, y, box_width, box_height)  ▷ Store the coordinates of the bounding boxes
end for
for i = 1 to NumberOfBounding_boxes do
    x, y, w, h ← bounding_boxes[i]
    cropped_image = image[y : y + h, x : x + w]
end for
```

Algorithm 3 Medical Image Processing and Text Retrieval

Input: PrescriptionPartImages, image_files[]
Output: text_file, text

```
for image_file = 1 to NumberOfimage_files do
    set_number_match = image_file
    if set_number_match then
        set_number = set_number_match.group(1)
    end if
    image_path = image_files
    img = image_path
    fig = figsize[10,10]
    height, width, channel ← img.shape
    text = OCR.image_to_string(img)
end for
```

4.7. Structured medicine data for further analysis

In this phase, the proposed system structures the medicine-related data of the text files obtained from the localized images and stores it in a central database. If the texts are arranged in multiple lines in the text files, first the proposed system brings all the lines to one line. These text files contain multiple data like names of the medicines, weight (mg, gram), dosage, and various instructional data about medication. Then, the proposed system uses regular expressions to structure the data according to the name of the drug, weight (mg or gram), and dosage (1+1+0) writing format like "X+X+X" for dose, where X represents a number. Then the proposed system tries to match the data in all the text files with the regular expression pattern. After processing the data of all the text files, the algorithm creates columns for medicine, weight, dosages, and rest of data using the panda's data frame and stores the related data in the center database. This algorithm efficiently structures the information related to medicine and organizes the data for later use, analysis, storage and later updating.

Fig. 9 shows that the proposed system structures the text data obtained from the ROI localized images and then stores the data in the central database. The process of structuring medicine data from prescription images is detailed in Algorithm 4.

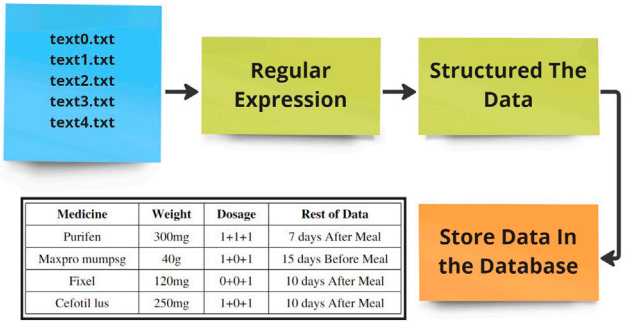


Fig. 9. Process of structuring the medicine-related data.

Algorithm 4 Medicine Data Extraction and Structuring

Input: PrescriptionData, Data[]
Output: ClassifyPrescriptionData, df[]

```
regular_expressions = {medicine names, weights, and dosages}
medicine_pattern = medicine names
weights_pattern = weights
dosages_pattern = dosages
for i = 1 to NumberOfLines do
    if line = value then
        if line_value = medicine_pattern then
            medicine_name ← medicine_value
            dosage ← dosages_value
            weight ← weights_value
            rest_of_data ← Extract the rest of the data
        end if
    end if
end for
df[] ← {medicine_name, weight, dosage, rest_of_data}
```

4.8. Multiclass ROI localization

In the previous processes, the proposed system basically imagined the medicine-related data as ROI and detected only those parts from the prescription image and then converted the medicine-related data to text with the help of OCR and kept the medicine-related data in a structured format. However, the prescription contains much more data like symptom name, test name, etc. The proposed system also applies this process to other data, such as localizing drug-related data as ROIs of one class and localizing the rest of the data symptom names, suggestions (test names, or others) as ROIs of another class. Then extract all the data and save the data in a structured way for later use. Fig. 10 shows that the proposed system can also identify other data (symptoms, test names, advice, etc.) from the prescription image extract the data if necessary, and store it in a central database in a structured format.

4.9. Web scraping for medication information extraction

The proposed system uses a Python script to scrape medication brand names, generic names, and indication (disease and symptom information) details from the 'medex.com.bd' website using BeautifulSoup libraries. It defines a function scrape_page to extract data from a single page, then iterates through multiple pages, and stores the data in a pandas DataFrame. The resulting data frame is saved as a CSV file.

Fig. 11 shows the web scraping process to extract medication data from the website, and save the data in a CSV file. Mainly this figure showcases a typical web scraping workflow for data collection. The medication Data extraction process is detailed in Algorithm 5.

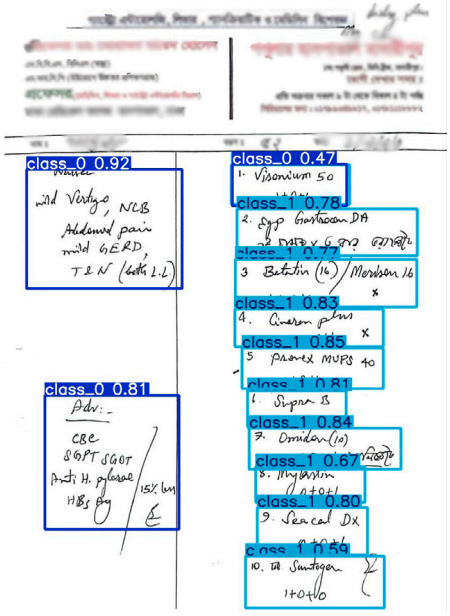


Fig. 10. Overview of multiclass ROI detection.

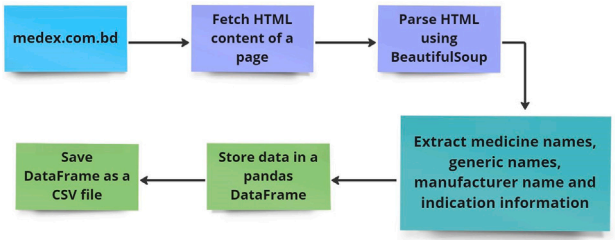


Fig. 11. Web scraping workflow for data collection.

Algorithm 5 Web-Based Medicine Data Scraper

Input: webpageText, webpage[]
Output: page_data[]

```
brand_links = soup.find_all('a', class)
for link = 1 to NumberOfbrand_links do
    name_div = link.find('div', class)                ▷ Extract medicine name
    name_text[] = name_div.get_text()
    all_divs[] = link.find_all('div', class)          ▷ Extract generic name
    details_text[] = all_divs[2].get_text()
    page_data[] ← {name_text[], details_text[]}       ▷ Store medicine and generic name
end for
```

4.10. Medicine name spell correction

The process is called Spell Checking and Correction. It corrects misspelled medicine names. The process of spell-checking and correcting medicine names uses two methods the Spello model and the Levnshtein edit distance algorithm.

4.10.1. Spello model for spell correction

The proposed system uses the Spello spell correction model to correct the misspelled words. Spello spell correction mainly uses two models "Phoneme", and "Symspell" in its backend to correct the misspelled words [57]. Symspell finds the correct word based on how close the misspelled word is to the existing dictionary word and minimum edit needs to be done to correct the misspelled word. Phoneme finds the correct word based on how words sound. Those two models work

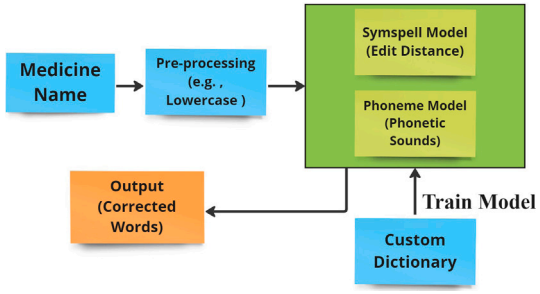


Fig. 12. Spello model workflow.

together in the Spello model backend to find the best correction for misspelled words.

Fig. 12 shows the Spello spell correction model Model workflow and which steps it follows to correct the misspelled words.

Now train the Spell Correction Model Spello using the medicine names data obtained from the medicine.txt file and save the trained model in a specified location. Then the code applies the trained spell correction spello model and retrieves the corrected name ('Fixal', 'Napa', and 'Purifen') from the model's output. The correct names are then updated in the structured medicine database and stored in the corrected names list for next use. The process of correcting the spelling of medicine names is detailed in Algorithm 6.

Algorithm 6 Medicine Name Spell Correction

Input: medicine_names[]
Output: corrected_medicine_names[]

```
LoadData = medicine_name.txt        ▷ Load medicine names dataset
CleanData = LoadData                ▷ Strip whitespace from each line
InitializeModel = Initialize the SpellCorrectionModel instance
TrainModel = CleanData
SaveModel = TrainModel
LoadModel = SaveModel
DefineMedicineNames = medicine_names[]
CorrectNames = Initialize an empty list for corrected_medicine_names[]
for i = 1 to len(medicine_names) do
    corrected_name ← Correct the spelling of medicine_names[i]
    corrected_medicine_names.append(corrected_name)
end for
```

4.10.2. Levnshtein edit distance algorithm for spell correction

The Spello Spell correction model corrects the wrong medicine names. But it has some drawbacks. If the English spelling of a wrong medicine name matches with another correct English spelling, then the Spello model considers it as the correct spelling. So the proposed system uses the Levnshtein edit distance algorithm to solve these drawbacks. Levenshtein Distance measures the minimum number of single-character edits (substitutions, insertions, deletions) required to change one word to another [58]. The Levenshtein distance between two strings a and b is given by:

$$lev_{a,b}(i,j) = \begin{cases} \max(i,j) & if\ \min(i,j) = 0 \\ \min \begin{cases} lev_{a,b}(i-1,j)+1 \\ lev_{a,b}(i,j-1)+1 \\ lev_{a,b}(i-1,j-1)+1 \text{ if } a_i \neq b_j \end{cases} & otherwise. \end{cases} \quad (2)$$

$lev_{a,b}(i,j)$ is the Levenshtein distance between the first i characters of string a and the first j characters of string b.
a_i is the ith character of string a.
b_j is the jth character of string b.

4.11. Extracting generic names, indication details, and related medication information

In this process, the proposed system extracts the medicine name and generic name assuming their positions in the columns. We store this data in a dictionary medicine_dict where the medicine name is the key and the generic name is the value. Next, the proposed system defines a list of medicine names to search for generic names - 'Fixal', 'Napa', and 'Purifen'. The proposed system creates an empty list of generic names to store the corresponding generic names. Then the proposed system iterates over each medicine name in medicine_names, checks if it exists in medicine_dict, and if found, the proposed system retrieves the generic name and adds it to the generic_names list. If a generic name is not found for a medicine name, the proposed system prints a message indicating so. This process demonstrates how to extract and match generic names for given medicine names from a dataset and also gives available brands of medicine names with their company names. Finally, the data will be stored in the central database. The same process is applied to extract and match Indications details for given generic names from a new dataset. The process of extracting generic names from medicine names is explained in Algorithm 7.

Algorithm 7 Extract Generic Names from Medicine Names

Input: CSVFilePath, MedicineNames[]
Output: GenericNames[]

```
csv_file ← csv.DictReader(file)
medicine_dict ← {}
for row = 1 to len(csv_file) do
    generic_name ← row["Generic_Name"]
    disease_name ← row["Additional_Info"]
    medicine_dict[row["Medicine_Name"]] ← generic_name
end for
for name = 1 to len(MedicineNames) do
    if name in medicine_dict then
        generic_names[name] ← medicine_dict[name]
    end if
end for
```

5. Evaluation and results

The proposed system uses YOLO ROI detection models (YOLOv7, YOLOv7 Tiny, YOLOv8, and YOLOv5) to detect the ROI parts from the prescription image. After that, the system extracts text from ROI localization parts from the prescription image using EasyOCR or Tressract OCR and stores the extracted text in a structured format. The system also uses spelling correction methods (Spello model, Levenshtein edit distance algorithm) to correct any errors in the extracted medicine names. In addition, it extracts medication information such as the name of the medicine, the company name, the generic name, and the indication details.

5.1. ROI detection and localization

In the proposed work, the system uses the YOLOv7, YOLOv7 Tiny, YOLOv8, and YOLOv5 models to detect or localize our prescription ROI. To select the best detection or localized model and evaluate the result we use the mean average precision matrix or confusion matrix. The more accurate the comparison, the better the detection accuracy of the model.

Average precision (AP) : The higher the value of Average precision, the better the performance of the model. Here Eq. (3) shows how average precision is calculated to measure the model performance.

$$\text{AP} = \frac{1}{11} \sum_{R_i} P(R_i) \quad (3)$$

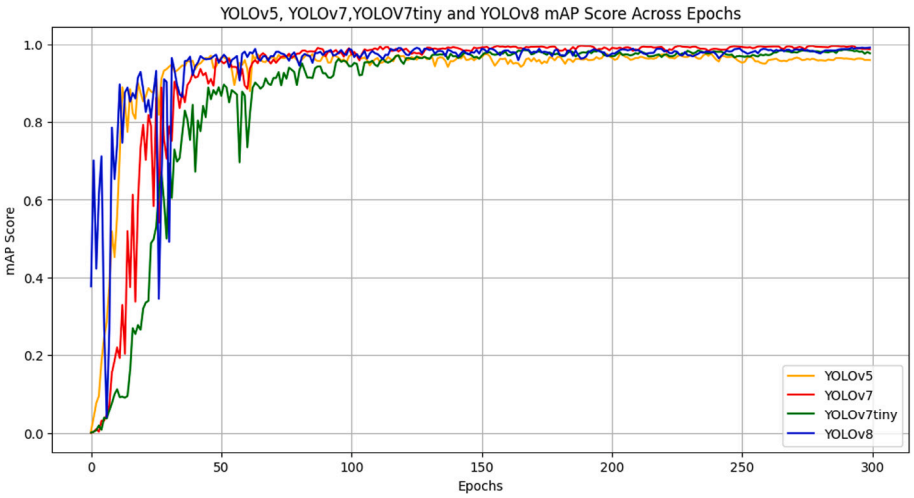


Fig. 13. Performance of the YOLO detection models in terms of mAP/epoch.

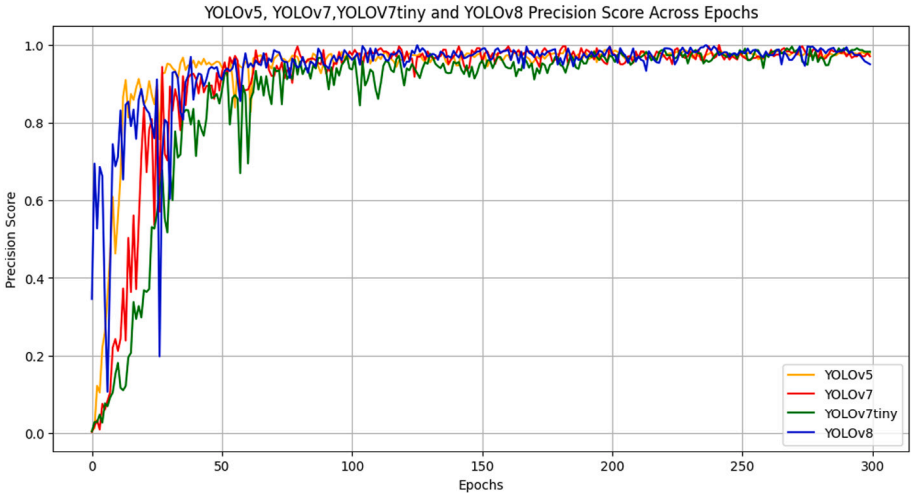


Fig. 14. Performance of the YOLO detection models in terms of precision/epoch.

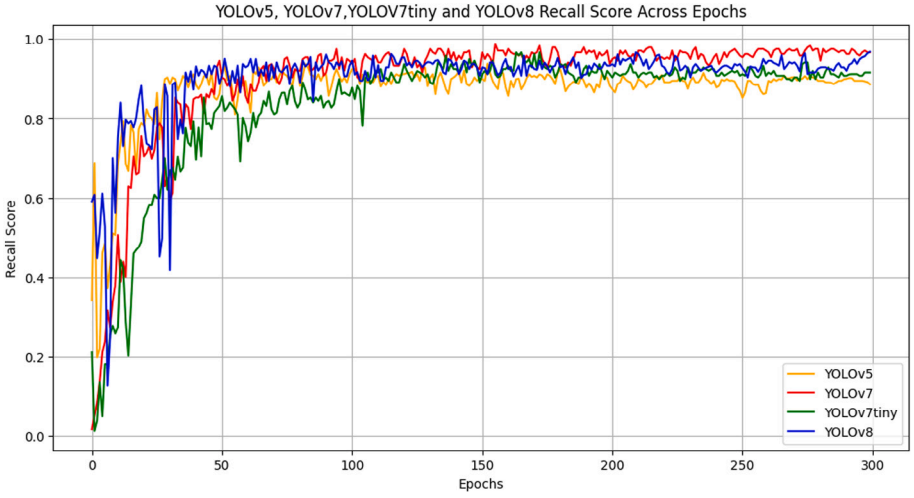


Fig. 15. Performance of the YOLO detection models in terms of recall/epoch.

Table 4
Performance of the models with validation data.

Model	YOLOv5	YOLOv7tiny	YOLOv7	YOLOv8
Precision	97.7	97.2	98.2	98.2
Recall	90.0	93.6	93.6	93.5
mAP@.5	96.1	97.6	98.6	98.5
Epochs	300	300	300	300

AP = Average Precision
P = Precision
R = Recall Value of ith Step

Mean average precision(mAP) : In object detection, overall accuracy is typically represented by the mean Average Precision (mAP). Unlike simple accuracy metrics, mAP considers both the localization accuracy (how well the predicted bounding boxes overlap with the ground truth) and the classification accuracy (whether the ROI is correctly classified). It aggregates performance across different IoU thresholds, providing a more comprehensive evaluation. A higher mAP score indicates improved accuracy in detecting Regions of Interest (ROIs) in prescription images. Here Eq. (4) shows how we calculate the mAP value to measure the model accuracy.

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i \tag{4}$$

mAP = Mean average precision
N = number of classes
AP_i = ith stage Average Precision Value

The proposed system needs a confusion matrix to calculate the mAP value. With the help of the confusion matrix, the system needs four elements for ROI part detection or localization of prescription images. True Positives (TP) occur when the model correctly detects ROI in prescription images, while True Negatives (TN) mean the model correctly identifies no ROI where none exists. False Positives (FP) happen when the model mistakenly detects ROI where there is none, and False Negatives (FN) occur when the model fails to detect ROI that is actually present.

Precision means how accurately the model has identified the ROIs of the prescription image. If the precision value is high, it means that the model has accurately detected almost all the ROIs.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \tag{5}$$

Recall means how accurately the model has been able to detect the original ROIs of the prescription image. A high Recall value means that the model is detecting all the prescription ROIs are real ROIs.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \tag{6}$$

F1 Score combines the precision and recall values into a single value. A high f1-score means that the model is detecting all the prescription ROIs with minimum false ROI and less number of missed ROIs.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

The Intersection over Union (IoU) process evaluates model accuracy by measuring how closely the predicted region matches the actual region, with predictions considered correct if IoU is greater than 0.5. To reduce errors, only predictions with over 50% confidence are included, ensuring more accurate evaluations.

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{8}$$

Table 5
Performance of the models with testing data.

Model	YOLOv5	YOLOv7tiny	YOLOv7	YOLOv8
Precision	99.5	97.2	1	99.4
Recall	97.4	93.6	99.5	98.5
mAP@.5	99.3	97.6	99.6	99.3
Epochs	300	300	300	300

The proposed system collects almost 400 prescription images. Then it divides the dataset into the train(70%), test(15%), and validation(15%) split to train the YOLO models. The proposed system rans 300 epochs to train the model. To evaluate the YOLO detection models' performance or how accurately models detect and localize ROI from prescription images the system uses precision, recall, and mAP@.5 parameters to find the best YOLO detection model.

Fig. 13 shows that the higher the mAP score of the model, the better the model. The proposed System runs 300 epochs to train the YOLO ROI detection models. It shows the mAP results for each epoch which helps measure the accuracy of the models.

Fig. 14 shows that the higher the precision score of the model, the better the model can detect the ROIs of the prescription image. It shows the precision results for each epoch which helps to measure how better the YOLO ROI detection models can detect ROIs in the prescription images.

Fig. 15 shows that the higher the recall score of the model, the more accurately the model can detect the ROIs of the prescription image. It shows the recall results for each epoch which helps to measure how accurately the YOLO ROI detection models can detect ROIs in the prescription images.

Table 4 shows the YOLOv7, YOLOv7 Tiny, YOLOv8, and YOLOv5 ROI detection models validation results using precision, recall, and mAP@.5 values.

Table 5 shows the YOLOv7, YOLOv7 Tiny, YOLOv8, and YOLOv5 ROI detection models testing results using precision, recall, and mAP@.5 values to find out the best model to detect ROI parts from our prescription images.

5.2. Optical character recognition

During the research period, the proposed system selected the two best open-source multiple languages OCRs available in the market, EasyOCR and Tesseract OCR.

Table 6 shows examples of EasyOCR and Tesseract OCR text extraction output. These OCRs extract text from Bangla and English printed images accurately. Also, it can extract text from English handwritten images but the limitation is noticeable when it comes to cursive handwriting. However, it cannot extract text from handwritten Bangla images. After analysis, the system shows that EasyOCR performs better than Tesseract OCR. One reason for this difference is EasyOCR's use of a deep learning approach that is essential for real-time object recognition and data analysis. Deep learning allows EasyOCR to adapt to different fonts, styles, and languages, increasing its effectiveness. So the system uses EasyOCR to extract text from images. After analyzing some printed Bangla and English prescription text recognition accuracy considering ROI parts from the prescriptions system find out that EasyOCR text recognition accuracy is more than 94% in terms of printed prescriptions. EasyOCR cannot perform well in English cursive handwritten text. On the other hand, if researchers use some paid OCR like Microsoft Azure AI Vision OCR and Google Cloud Vision OCR output for cursive English handwritten data extraction, then the text extraction accuracy comes close to 98% which is shown in Table 7. Also these OCRs cannot extract text from handwritten Bangla images.

To illustrate the accuracy of text extraction using OCR shows in Tables 6 and 7, we calculate the similarity between the original text

Table 6
Sample of some EasyOCR and Tesseract OCR text extraction output.

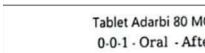
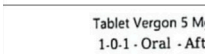
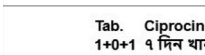
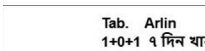
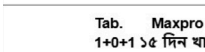
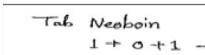
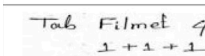
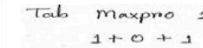
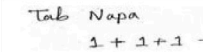
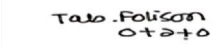
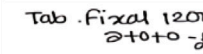
Input Image	EasyOCR Output	Accuracy (%)	Tesseract OCR Output	Accuracy (%)
	Tablet Bislol 5 Mg Tab 1-0-0 Oral After Meal Cont	98.99	Tablet Bislol 5 Mg Tab 1-0-0 Oral After Meal Cont	98.99
	Tablet Adarbi 80 MG 0-0-1 Oral After Meal Cont	93.48	Tablet Adarbi 80 MG 0-0-1 Oral After Meal Cont	93.48
	Tablet Vergon 5 Mg Tab 1-0-1-Oral-After Meal.Cont	97.96	Tablet Vergon 5 Mg Tab 1-0-1 Oral After Meal Cont	93.48
	Tab. Ciprocin 500mg 1+0+1 ৭ দিন খাবার পর খাবেন	100	Tab. Ciprocin 500rg 1+0+1 দিন খাবার পর খাবেন	95.45
	Tab. Arlin 500mg 1+0+1 ৭ দিন খাবার পর খাবেন	100	Tab. Arlin 500mg 1+0+1 এদিন খাবার পর খাবেন	96.47
	Tab. Maxpro 40mg 1+0+1 ১৫ দিন খাবার পর খাবেন	100	Tab. Maxpro 40mg 1+0+1 ১০ দিন খাবার পর খাবেন	97.67
	Tab Neoboin 4+0+1-1 month	96.00	Tab Neaboin L+o+4 201414 Month	78.43
	TabFilmet 1+1+1 2 days	79.25	Tal Filmet Goo m fei eh +4 days	62.30
	Tab Maxpro 20 mq 1+0++ 4 3 datb	75.86	Tab Moxpro 20 4+0 +4 Sdoys	73.08
	Tb Napa 1+1+1 Aflen mel	80.85	Tob Napa 1+ 1e1 Affe meal	77.55
	Tab .Foliton Ota+0 = J7Tt	61.22	Tato .Folicon O+d+0 - MNS	66.67
	Tab.Fixol 120mq Dt+0 Jmm nira	52.31	Tab-Fixat 120m 4o1o prem	53.33

Table 7
Sample of some paid OCR text extraction output.

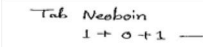
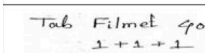
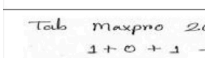
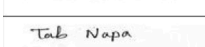
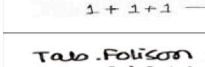
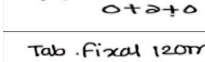
Input Image	Microsoft Azure AI Vision OCR	Accuracy (%)	Google Cloud Vision OCR Output	Accuracy (%)
	Tab Neoboin 1+0+1-1 month	100	Tab Neoboin 1+0+1-1 month	100
	Tab Filmet 400 mg 1+1+1-3 days	100	Tab Filmet 400 mg 1+1+1-3 days	100
	Tab Maxpro 20mg 1+0+1-3days	100	Tab maxpro 20mg 1+0+1 3days	100
	Tab Napa 1+1+1-Aftermeal	100	Tab Napa 1+1+1-Aftermeal	100
	Tab.Folison O+2+0-roman	72.34	Tab.Folison otato-mor	66.67
	Tab.Fixal 120mg 2+0+0-5mommmon	63.64	Tab.Fixal 120mg er*-1+0onmm	60.32

Table 8
Example of prescription image ROI-localized parts medicine-related Data.

Medicine-related Data
Tab Purifen 300 mg 1+1+1 7 days After Meal
Maxpro mumps 40 g 1+0+1 15 days Before Meal
Tab Fixel 120 mg 0+0+1 10 days After Meal
Tab. Cefotil lus 250 mg 1+0+1 7 days After Meal

Table 9
Storing medicine-related data in structured format.

Medicine	Weight	Dosage	Rest of data
Purifen	300 mg	1+1+1	7 days After Meal
Maxpro mumpsg	40 g	1+0+1	15 days Before Meal
Fixel	120 mg	0+0+1	10 days After Meal
Cefotil lus	250 mg	1+0+1	10 days After Meal

Table 10
Example of some misspelled medicine names spell correction processes.

Misspelled medicine name	Correct medicine name	Spello model	Levenshtein algorithm
nopa	napa	nop	napa
cefotil lus	cefotil plus	cefotil plus	cefotil plus
maxpro mumps	maxpro mups	maxpro mups	maxpro mups
mix	afix	mix	afix
in call extra	on call extra	in call extra	on call extra
ab derm cold cream corps	abc derm cold cream corps	ab derm cold cream corps	abc derm cold cream corps
one glod	one gold	one glod	one gold
porifap	paricap	soricap	paricap
onecon	onicon	onicon	onicon
ads	pds	ds	pds

Table 11
Medicine related data.

Medicine name	Weight	Dosage	Rest of data	Corrected name
Purifen	300 mg	1+1+1	7 days After Meal	Purifen
Maxpro mumpsg	40 g	1+0+1	15 days Before Meal	Maxpro mumps
Fixel	120 mg	0+0+1	10 days After Meal	Fixel
Cefotil lus	250 mg	1+0+1	10 days After Meal	Cefotil plus

and the OCR-extracted text. Here Eq. (9) shows how we calculate the OCR text extraction accuracy.

$$\text{Accuracy} = \frac{2 \times \text{Number of Matching Characters}}{\text{Sum of original and OCR-extracted text length}} \times 100 \quad (9)$$

5.3. Structured medicine data analysis

The proposed system organizes medicine-related data obtained from prescription image ROI segments after processing text extraction from the images. After processing all the data, store the data in the central database.

Table 8 shows the medicine-related data that the system extracted from ROI-localized images of prescriptions. Now system needs to structure those data.

Table 9 shows how the system stores the medicine-related data in a structured format for easy use in future analysis and extracts more medication information.

5.4. Spell correction

The proposed system checks the spelling of the medicine name from the medicine detail structured database and if there are any spelling errors, the system will correct them using spelling correction methods (Spello model or Levenshtein edit distance algorithm). The Spello model has some drawbacks. If the English spelling of an incorrect drug name matches another correct English spelling, the Spello model treats it as the correct spelling. So the system use the Levnshtein edit distance algorithm to solve these problems. The system evaluates the accuracy of spelling correction using the Spello model and the Levenshtein edit distance algorithm on a list of 100 misspelled medicine names. The system achieved an accuracy of about 95% mainly with the Spello model and the Levenshtein algorithm showed an accuracy of about 96%.

Table 10 illustrates and compares the Spello model and the Levenshtein edit distance algorithm in correcting misspelled drug names. This sample example shows how the Levenshtein algorithm is more accurate in dealing with misspellings of medicine names.

5.5. Final output summary

The system shows how the information extracted from the ROI localization part of the prescription image is stored in the central database using three tables, and each table is designed for different medicine names information organization, and management.

Table 11 shows how the central database stores basic information about each medicine, including the medicine's revised name, weight, dosage, and instructions for use. The system then captures and organizes this information after correcting misspelled medicine names. This information in structured data makes it easy to use and verify any basic data for each medicine.

Table 12 shows how more medicine-related information for each medicine is stored in the central database, where medicine names appear under its generic name with other brand name alternatives and their indications details. This information is collected from various sources which helps the central database with this detailed information as well as giving a more precise picture of each medicine and its uses.

Table 13 links the corrected medicine name with the medicine manufacturing company. This table helps maintain a clear record of which companies are responsible for each medicine, ensuring better management in the centralized system.

A comparative statistics of researchers findings and the proposed work is provided in Table 14. The studies are categorized based on reference, methodology, dataset, and performance, offering a comparative overview of the approaches and results in prescription data extraction.

6. Conclusion

The authors propose an EHR system that uses technologies like computer vision, OCR, machine learning, and web scraping to extract and structure medical data from handwritten and physical prescriptions. The system assumes standard prescription formats and high-quality images but challenges include handwriting variability, image quality, and non-standard prescription layouts. Integrating pre-processing techniques and fine-tuning models on diverse datasets can improve robustness. Despite limitations, the system effectively structures medical data, enriches it with medication details, and holds potential to streamline healthcare management, reduce manual effort, and enhance patient care. This research enhances patient data access, improves diagnosis and treatment accuracy, and enhances patient care through secure digital storage. It benefits healthcare providers, researchers, governments, and policymakers by enabling faster data access, advancing medical extraction technologies, and promoting integrated solutions. The findings will be disseminated through academic journals, conferences, workshops, webinars, collaborations, and public outreach. The

Table 12
Extra medication information data.

Corrected medicine name	Generic name	Alternative brands name	Indication details
Purifen	Dexibuprofen	Artoflex, Dex-i, Dip, Fenexi, Inflam-D	Purifen is indicated for osteoarthritis, rheumatological disorders, painful menstruation, headaches, fever, and mild to moderate pain, and as an adjuvant for headaches with colds and influenza.
Maxpro mumps	Esomeprazole	EMT MUPS, Esofour MUPS, Esomups	Maxpro MUPS tablet is indicated in: Gastro-esophageal Reflux Disease (GERD), Risk reduction in NSAID-associated gastric ulcer, H. pylori eradication (Triple therapy), Zollinger-Ellison syndrome, and idiopathic hypersecretion.
Fixel	Fexofenadine Hydrochloride	Adifex, Afexa, Alafree, Alanil	Fixel is indicated for the relief of symptoms associated with seasonal and perennial allergic rhinitis, in adults and children 12 years of age and over. Symptoms treated effectively include sneezing, rhinorrhea, lacrimation, itchy, red eyes, and itchy nose/palate/throat. Fixel improves health-related quality of life and work/activity productivity.
Cefotil plus	Cefuroxime Axetil + Clavulanic Acid	Axeclav, Axicef Plus, Axim Cv	Cefotil Plus is indicated for treating various bacterial infections, including respiratory tract infections, skin infections, urinary tract infections, bone and joint infections, gonorrhea, Lyme disease, septicemia, meningitis, and for switching therapy from injectable to oral treatment.

Table 13
Medicine name and company name data.

Medicine name	Company name
Purifen	Incepta Pharmaceuticals Ltd.
Maxpro Mumps	Renata Limited
Fixel	Sharif Pharmaceuticals Ltd.
Cefotil Plus	Square Pharmaceuticals PLC

first three months of the prescription digitization plan focus on expanding the dataset, improving technologies, and validating. The next six months will build and test a prototype, implement it in healthcare, and integrate systems. After a year, the system will be widely accessible, secure, user-educated, and continuously updated. The proposed method has applications in various sectors beyond healthcare, including financial document analysis, logistics, manufacturing quality control, and security and surveillance applications. It accurately identifies image regions, extracts text, corrects errors, and structures the data, making it valuable for tasks like data entry, accounting, auditing, package tracking and delivery, inspecting shipping documents and manufacturing defects, historical document analysis, and license plate detection.

7. Future work

In the future, the author plans to implement federated learning to enhance the security and privacy of users' data. Federated learning allows us to train machine learning models across multiple decentralized devices or servers holding local data samples, without exchanging them. This approach ensures that personal and sensitive data remains on local devices, significantly enhancing data privacy and protection while still enabling robust model training. Also, we are looking forward to cross-checking medication symptoms. When a generic name is identified from a scanned prescription, the symptoms associated with the prescribed medication will be cross-checked with the symptoms described in the prescription. This cross-referencing will help ensure the accuracy and appropriateness of the prescribed treatment, providing an additional layer of verification. The study used primary prescription data from friends and relatives, with consent and confidentiality. Future research will use anonymized data from local hospitals and clinics, expanding the dataset with more handwriting styles and prescription formats. The results of this quantitative evaluation will be tested on the primary dataset and new hospital-acquired data by comparing precision, accuracy, recall analysis, F1 score, and ROI detection. Future EHR systems should adopt personalized training to address variations in doctors' handwriting by fine-tuning OCR models with individual samples. By incorporating continuous learning, NLP, and interactive feedback, these systems can enhance accuracy and ensure reliable extraction of data from handwritten records. Additionally, in the future, we will focus on implementing the system in real-world existing healthcare settings, and data privacy. We will also explore scalability by ensuring the system adapts to different healthcare environments and technological infrastructures. This study develops an EHR system to extract prescription data. This technology can create more functions such as drug interaction testing, allergy alerts, and drug usage tracking. The technology will also manage other patient information, such as lab results, diagnoses, and medical history, creating one place for all patient care information. This will make data accessible, improve doctor decisions, and keep patients safer.

CRediT authorship contribution statement

Apurba Datta: Writing – original draft, Formal analysis, Conceptualization. **Md. Mehedi Hasan:** Methodology, Investigation. **Niaz Mahmud:** Writing – original draft, Visualization. **Bilkis Jamal Ferdosi:** Supervision. **Rafiqul Islam:** Writing – review & editing, Conceptualization. **Ziaur Rahman:** Writing – review & editing. **Sk. Tanzir Mehedi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for *Computers in Biology and Medicine* and was not involved in the editorial review or the decision to publish this article.

Table 14
Comparative analysis of reviewed literature and proposed work on prescription data extraction.

Ref.	Methodology	Dataset	Accuracy
[32]	Line/word segmentation, SVM	50 prescription samples	Line: 95%, Word: 92%, Recognition: 85%
[33]	YOLOv4, CRNN-CTC	112 prescriptions	98% accuracy
[34]	Weakly supervised method, OCR, CTC	9645 prescriptions, 90k medical names	Not specified
[41]	NLP	800k medical statements	95% accuracy
[43]	LSTM, RSS augmentation	17 431 medical terms	93% accuracy
[35]	OCR, BERT, VGG-16	Own dataset	94.44% image classification, 99.9% term classification
[36]	CRNN	Dataset with 66 classes (alphanumeric, punctuation)	98% accuracy
[37]	OCR, CNN, RNN, LSTM, Fuzzy search	Not specified	Not specified
Proposed	YOLO ROI detection models, OCR (Tesseract OCR, and EasyOCR), Regular expression, Spello model, Levenshtein Edit Distance Algorithm, web scraping technique	Dataset includes 400 prescription images with 2200 ROI coordinates, 15,106 medication entries with brand and generic names, and 1628 entries linking generic medications to symptoms or diseases.	Prescription images ROI Detection accuracy: 99.6%, Medicine Name Spell Correction accuracy: 96%

References

[1] K. Arnold, G.A. Mugisha, F.-M.U.S. Imanirakiza, C. Muhumuza, J.N. Bukenya, Development of an E-Health System for Improving Health-Care Access in Developing Countries, in: Lecture Notes in Networks and Systems, Springer International Publishing, Cham, 2022, pp. 607–616.

[2] UNDP, Covid-19: A reality check for Bangladesh's healthcare system, 2020, URL: https://www.undp.org/bangladesh/stories/covid-19-reality-check-bangladeshs-healthcare-system (Accessed 1 March 2024).

[3] A. Islam, T. Biswas, Health system in Bangladesh: Challenges and opportunities, Am. J. Heal. Res. 2 (2014) 366, http://dx.doi.org/10.11648/j.ajhr.20140206.18.

[4] M. Shahen, R. Ahmed, M. Islam, Challenges for health care services in Bangladesh: An overview, IOSR J. Nurs. Heal. Sci. (IOSR-JNHS) 9 (2022) 13–24, http://dx.doi.org/10.9790/1959-0901011324.

[5] T.O. Edoh, Challenges facing healthcare delivery systems in low-, middle-, and high-income countries, in: Research Anthology on Public Health Services, Policies, and Education, IGI Global, 2021, pp. 299–352.

[6] World Health Organization, Global Spending on Health: Coping with the Pandemic, World Health Organization, Genève, Switzerland, 2024.

[7] National Budget, Healthcare spending up by just 3.2% even as budget sees 12.34% rise, 2023, https://www.thedailystar.net/special-events/national-budget-2023-24/news/healthcare-spending-just-32-even-budget-sees-1234-rise-3335226 (Accessed 19 April 2024).

[8] S. Hasan, S. Hossain, R. Mamun, D.S. Bhattacharyya, R. Ahmed, F. Sultana, R. Gazi, A. Adams, Medical record keeping practices among the urban poor in Bangladesh, 2016.

[9] J. Moini, O. Akinso, K. Ferdowsi, M. Moini, Medical record-keeping, in: Health Care Today in the United States, Elsevier, 2023, pp. 499–514, http://dx.doi.org/10.1016/B978-0-323-99038-7.00015-1, URL: https://linkinghub.elsevier.com/retrieve/pii/B9780323990387000151.

[10] W. Wang, D. Ferrari, G. Haddon-Hill, V. Curcin, Electronic health records as source of research data, in: Machine Learning for Brain Disorders, Springer US, New York, NY, 2023, pp. 331–354, http://dx.doi.org/10.1007/978-1-0716-3195-9_11.

[11] K. Luong, What is EHR? A comprehensive overview, 2024, https://www.ominext.com/en/blog/what-is-ehr-a-comprehensive-overview (Accessed 30 May 2024).

[12] K.A. Pereira, Toward enhanced healthcare efficiency: The impact of digitizing medical records, Rev. Sist. 14 (3) (2024) 709–714, http://dx.doi.org/10.56238/rcsv14n3-020.

[13] M. Nasim, S. Abdullah-Al-Noman, A. Ragib Hasan, A. Sattar, Digitalization and centralization of medical information and patient history in Bangladesh, in: 2022 6th International Conference on Computing Methodologies and Communication, ICCMC, IEEE, 2022, pp. 153–158, http://dx.doi.org/10.1109/ICCMC53470.2022.9753690.

[14] Adekunle Oyeyemi Adeniyi, Jeremiah Olawumi Arowoogun, Rawlings Chidi, Chioma Anthonia Okolo, Oloruntoba Babawarun, The impact of electronic health records on patient care and outcomes: A comprehensive review, World J. Adv. Res. Rev. 21 (2) (2024) 1446–1455, http://dx.doi.org/10.30574/wjarr.2024.21.2.0592.

[15] Rahaf Abdullah Alomair, Dr. Syed Arif Pasha, An emperical investigation into the impact of electronic health records on healthcare quality and patient safety, Open Access Public Heal. Heal. Adm. Rev. 2 (2) (2024) 30–38, http://dx.doi.org/10.59644/oaphhar.2(2).97.

[16] A.V. Razzano, A scoping review on the impacts of electronic health record systems on healthcare delivery, Int. J. Heal. Sci. Res. 14 (7) (2024) 276–282, http://dx.doi.org/10.52403/ijhsr.20240737.

[17] L. Slawomirski, L. Lindner, K. de Bienassis, P. Haywood, T.C.O. Hashiguchi, M. Steentjes, J. Oderkirk, Progress on implementing and using electronic health record systems: Developments in OECD countries as of 2021, Organ. Econ. Co-Oper. Dev. (OECD) (2023) URL: https://doi.org/10.1787/4f4ce846-en.

[18] J. Green, How much EHR costs and how to set your budget, 2024, https://www.ehrinpractice.com/ehr-cost-and-budget-guide.html (Accessed 25 May 2024).

[19] R. Slawter, How to establish a budget for an EHR system (with free budget template), 2024, https://www.tebra.com/theintake/practice-operations/patient-care/how-to-establish-a-budget-for-an-ehr-system-with-free-budget-template (Accessed 25 May 2024).

[20] F. Hossain, M.M. Bouh, M.M. Rahman, F. Shah, T. Mine, R. Islam, N. Nakashima, A. Ahmed, Status and challenges of medical history taking in Bangladesh and an affordable digital solution to tackle them, Appl. Syst. Innov. 7 (4) (2024) http://dx.doi.org/10.3390/asi7040069, URL: https://www.mdpi.com/2571-5577/7/4/69.

[21] E. Yadav, E.S. Singh, Centralized health coordination system for managing medical records, Int. J. Res. Appl. Sci. Eng. Technol. 12 (5) (2024) 4895–4904, http://dx.doi.org/10.22214/ijraset.2024.62687.

[22] P. Palma, M.A.-M. Molla, Prescription as usual, 2017, https://www.thedailystar.net/frontpage/prescription-usual-1368460 (Accessed 17 May 2024).

[23] S. Chaturvedi, What's wrong with doctors' handwriting? Natl. Med. J. India 31 (1) (2018) 47.

[24] M.A. Khan, H. Arif, V. Singh, D. Sharma, Decoding the script: A bi-LSTM based journey into doctor's handwriting for precise prescription understanding, in: 2023 International Conference on Smart Devices, ICSD, IEEE, 2024, pp. 1–6.

[25] S. Butala, A. Lad, H. Chheda, M. Bhat, A. Nimkar, Natural language parser for physician's handwritten prescription, in: 2020 International Conference on Emerging Trends in Information Technology and Engineering, Ic-ETITE, IEEE, 2020, pp. 1–7.

[26] Wikipedia contributors, Medical prescription, 2024, https://en.wikipedia.org/w/index.php?title=Medical_prescription&oldid=1242829998 (Accessed 20 December 2024).

[27] T. Mankash, V.S. Chaithanya Kota, A. De, P. Prakash, K. Jadhav, MIRAGE: Multimodal identification and recognition of annotations in Indian general prescriptions, 2024, http://dx.doi.org/10.48550/arXiv.2410.09729, arXiv.

[28] M. Haque, S.M. Barik, S. Bashar, R. Islam, S. Bashar, A. Kumar, A. Dipon, N. Kona, A. Chouduri, Errors, omissions and medication patterns of handwritten outpatient prescriptions in Bangladesh: A cross-sectional health survey, J. Appl. Pharm. Sci. 6 (2016) 042–046, http://dx.doi.org/10.7324/japs.2016.60608.

[29] A. Kiourtis, A. Mavrogiorgou, A. Menychtas, I. Maglogiannis, D. Kyriazis, Structurally mapping healthcare data to HL7 FHIR through ontology alignment, J. Med. Syst. 43 (3) (2019) 62.

[30] A. Mavrogiorgou, A. Kiourtis, M. Touloupou, E. Kapassa, D. Kyriazis, Internet of medical things (IoMT): Acquiring and transforming data into HL7 FHIR through 5G network slicing, Emerg. Sci. J. 3 (2) (2019) 64.

[31] R. Saripalle, C. Runyan, M. Russell, Using HL7 FHIR to achieve interoper- ability in patient health records, J. Biomed. Inform. 94 (2019).

[32] P.S. Dhande, Kharat, Reena, Character recognition for cursive english handwriting to recognize medicine name from doctor's prescription, in: 2017 International Conference on Computing, Communication, Control and Automation, ICCUBEA, IEEE, Pune, 2017, pp. 1–5, http://dx.doi.org/10.1109/ICCUBEA.2017.8463842, URL: https://ieeexplore.ieee.org/document/8463842/.

[33] Edlabadkar, S. Ravindra, Joshi, Pranjali, Accurate medicine recognition and audio conversion for handwritten prescriptions, in: 2024 International Conference on Emerging Smart Computing and Informatics, ESCI, IEEE, Pune, India, 2024, pp. 01–06, http://dx.doi.org/10.1109/ESCI59607.2024.10497304, URL: https://ieeexplore.ieee.org/document/10497304/.

[34] S. Paul, G. Madan, A. Mishra, N. Hegde, P. Kumar, G. Aggarwal, Weakly supervised information extraction from inscrutable handwritten document images, in: G.A. Fink, R. Jain, K. Kise, R. Zanibbi (Eds.), Document Analysis and Recognition - ICDAR 2023, vol. 14190, Springer Nature Switzerland, Cham, 2023, pp. 445–463, http://dx.doi.org/10.1007/978-3-031-41685-9_28, Series Title: Lecture Notes in Computer Science URL: https://link.springer.com/10.1007/978-3-031-41685-9_28.

[35] A.M. Sakib, B. Jamal Ferdosi, S. Jahan, K. Jashim, Medical text extraction and classification from prescription images, in: 2022 25th International Conference on Computer and Information Technology, ICCIT, IEEE, Cox's Bazar, Bangladesh, 2022, pp. 472–477, http://dx.doi.org/10.1109/ICCIT57492.2022.10055123, URL: https://ieeexplore.ieee.org/document/10055123/.

[36] R. Achkar, K. Ghayad, R. Haidar, S. Saleh, R. Al Hajj, Medical handwritten prescription recognition using CRNN, in: 2019 International Conference on Computer, Information and Telecommunication Systems, CITS, IEEE, Beijing, China, 2019, pp. 1–5, http://dx.doi.org/10.1109/CITS.2019.8862004, URL: https://ieeexplore.ieee.org/document/8862004/.

[37] Pavithiran, S. Padmanabhan, N. Divya, V. Aswathy, P.I. Jerusha, B. Chandar, Doctor's handwritten prescription recognition system in multi-language using deep learning, in: 2022 International Conference on Power, Energy, Control and Transmission Systems, ICPECTS, IEEE, Chennai, India, 2022, pp. 1–5, http://dx.doi.org/10.1109/ICPECTS56089.2022.10047588, URL: https://ieeexplore.ieee.org/document/10047588/.

[38] Sarkar, Sayantan, Word spotting in cursive handwritten documents using modified character shape codes, in: N. Meghanathan, D. Nagamalai, N. Chaki (Eds.), Advances in Computing and Information Technology, vol. 178, Springer Berlin Heidelberg, Berlin, Heidelberg, 2013, pp. 269–278, http://dx.doi.org/10.1007/978-3-642-31600-5_27, Series Title: Advances in Intelligent Systems and Computing. URL: https://link.springer.com/10.1007/978-3-642-31600-5_27.

[39] P.P. Roy, A.K. Bhunia, A. Das, P. Dhar, U. Pal, Keyword spotting in doctor's handwriting on medical prescriptions, Expert Syst. Appl. 76 (2017) 113–128, http://dx.doi.org/10.1016/j.eswa.2017.01.027, URL: https://linkinghub.elsevier.com/retrieve/pii/S0957417417300337.

[40] K. Shanmugalingam, S. Sumathipala, C. Premachandra, Word level language identification of code mixing text in social media using NLP, in: 2018 3rd International Conference on Information Technology Research, ICITR, IEEE, Moratuwa, Sri Lanka, 2018, pp. 1–5, http://dx.doi.org/10.1109/ICITR.2018.8736127, URL: https://ieeexplore.ieee.org/document/8736127/.

[41] V. Carchiolo, A. Longheu, G. Reitano, L. Zagarella, Medical prescription classification: A NLP-based approach, in: 2019 Federated Conference on Computer Science and Information Systems, FedCSIS, 2019, pp. 605–609, URL: https://api.semanticscholar.org/CorpusID:202728713.

[42] I. Khandokar, M. Hasan, F. Ernawan, S. Islam, M.N. Kabir, Handwritten character recognition using convolutional neural network, J. Phys.: Conf. Ser. 1918 (4) (2021) 042152, http://dx.doi.org/10.1088/1742-6596/1918/4/042152, URL: https://iopscience.iop.org/article/10.1088/1742-6596/1918/4/042152.

[43] S. Tabassum, N. Abedin, M.M. Rahman, M.M. Rahman, M.T. Ahmed, R. Islam, A. Ahmed, An online cursive handwritten medical words recognition system for busy doctors in developing countries for ensuring efficient healthcare service delivery, Sci. Rep. 12 (1) (2022) 3601, http://dx.doi.org/10.1038/s41598-022-07571-z, URL: https://www.nature.com/articles/s41598-022-07571-z.

[44] K. Sharma, S.J. Kona, A. Jangwal, A. M, P. Rajabai C, D.R. Sona, Handwritten Digits and Optical Characters Recognition, Int. J. Recent. Innov. Trends Comput. Commun. 11 (4) (2023) 20–24, http://dx.doi.org/10.17762/ijritcc.v11i4.6376, URL: https://ijritcc.org/index.php/ijritcc/article/view/6376.

[45] D. Vedhaviyassh, R. Sudhan, G. Saranya, M. Safa, D. Arun, Comparative analysis of easyOCR and tesseractOCR for automatic license plate recognition using deep learning algorithm, in: 2022 6th International Conference on Electronics, Communication and Aerospace Technology, IEEE, Coimbatore, India, 2022, pp. 966–971, http://dx.doi.org/10.1109/ICECA55336.2022.10009215, URL: https://ieeexplore.ieee.org/document/10009215/.

[46] M.M. Yulianto, R. Arifudin, A. Alamsyah, Autocomplete and spell checking levenshtein distance algorithm to getting text suggest error data searching in library, Sci. J. Inform. 5 (1) (2018) 75, http://dx.doi.org/10.15294/sji.v5i1.14148, URL: https://journal.unnes.ac.id/nju/index.php/sji/article/view/67.

[47] S.A. Lete, C. Cavero, M. Lustrek, D. Kyriazis, A. Kiourtis, J. Mantas, L. Montandon, Interoperability techniques in CrowdHEALTH project: The terminology service, Acta Inf. Med. 27 (5) (2019) 355–361.

[48] E. Kouremenou, A. Kiourtis, D. Kyriazis, A data modeling process for achieving interoperability, in: IFMBE Proceedings, Springer Nature Switzerland, Cham, 2024, pp. 711–719.

[49] K. Mavrogiorgos, A. Kiourtis, A. Mavrogiorgou, S. Kleftakis, D. Kyriazis, A multilayer approach for data cleaning in the healthcare domain, in: Proceedings of the 2022 8th International Conference on Computing and Data Engineering, 2022, pp. 22–28, http://dx.doi.org/10.1145/3512850.3512856.

[50] P. Misra, A. Yadav, Impact of preprocessing methods on healthcare predictions, SSRN Electron. J. (2019) http://dx.doi.org/10.2139/ssrn.3349586.

[51] C.-Y. Wang, A. Bochkovskiy, H.-Y.M. Liao, YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors, 2022, arXiv preprint arXiv:2207.02696.

[52] G. Jocher, A. Chaurasia, J. Qiu, Ultralytics YOLOv8, 2023, URL: https://github.com/ultralytics/ultralytics.

[53] G. Jocher, Ultralytics YOLOv5, 2020, http://dx.doi.org/10.5281/zenodo.3908559, URL: https://github.com/ultralytics/yolov5.

[54] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: Unified, real-time object detection, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR, IEEE, Las Vegas, NV, USA, 2016, pp. 779–788, http://dx.doi.org/10.1109/CVPR.2016.91, URL: http://ieeexplore.ieee.org/document/7780460/.

[55] A. Mahajan, EasyOCR: A comprehensive guide - Aditya Mahajan, 2023, URL: https://medium.com/@adityamahajan.work/easyocr-a-comprehensive-guide-5ff1cb850168 (Accessed 10 May 2024).

[56] F. Zelic, OCR using pytesseract and OpenCV, 2023, URL: https://nanonets.com/blog/ocr-with-tesseract/ (Accessed 10 April 2024).

[57] S. Aman, R.S. Ruthvik, Spello: Fast and accurate spell correction library, 2020, URL: https://github.com/hellohaptik/spello (Accessed 11 April 2024).

[58] H. Patel, G. Ravichandran, Levenshtein algorithm, 2022, http://dx.doi.org/10.17613/z4b8-b588, URL: https://hcommons.org/deposits/objects/hc:46164/datastreams/CONTENT/content (Accessed 11 May 2024).



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